

Notes: Dilmé & Garrett (RES, 2026)

A Dynamic Theory of Random Price Discounts

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1 Big picture

- **Question.** Why do many retailers post a high regular price most of the time, interrupted by abrupt and randomly timed discounts, rather than using deterministic gradual markdowns?
- **Core mechanism.** A seller with commitment power faces forward-looking, risk-averse buyers arriving over time. Random discounts implement the allocation probabilities of the optimal static mechanism while shielding each buyer type from arrival-time pricing risk.
- **Empirical motivation.** Retail price paths often have two focal prices, abrupt discounts, and no obvious pre-sale collapse in purchases.
- **Static step.** Concavity of buyer utility can make random allocation optimal: some types receive the good with probability strictly between zero and one.
- **Dynamic step.** The seller implements those allocation probabilities with waiting times for random discounts.
- **Economic takeaway.** Random sales can be optimal intertemporal price discrimination when buyers are forward-looking but dislike price risk.

2 Model primitives and measurement

2.1 Seller, buyers, and types

There is a monopolist seller, zero marginal cost, and one unit demand. Buyer types are ordered $\theta_N > \dots > \theta_1 > 0$, with probabilities $\beta_n > 0$. If type θ_n buys at price p , the seller receives p and the buyer receives $v_n(p) = v(p; \theta_n)$.

$$v_n(\theta_n) = 0.$$

Interpretation. The normalization makes θ_n the maximum willingness to pay. Concavity of v_n makes price risk costly.

2.2 Preference restrictions

Condition A1 limits price sensitivity of higher types, and Condition A2 says higher types are weakly less risk averse. The main text also studies CARA utility:

$$v_n(p) = 1 - e^{-R(\theta_n - p)}.$$

Interpretation. These restrictions generate the single-crossing structure needed for the optimal mechanism.

2.3 Static mechanisms

A static mechanism assigns each type an allocation probability x_n and a deterministic price p_n conditional on receiving the good. Lemmas 1 and 2 imply deterministic payments and permit the optimal mechanism to be characterized by binding downward IC constraints.

2.4 Dynamic price-path mechanisms

A dynamic mechanism is a stochastic price process P_t and stopping times $\tau_{n,t}$. Incentive compatibility requires each type to prefer the prescribed stopping time to every alternative stopping time adapted to the observed price history.

3 Models and theoretical specifications

3.1 Optimal static mechanism

$$x_N^* = 1, \quad x_n^* v_n(p_n^*) = x_{n-1}^* v_n(p_{n-1}^*).$$

Interpretation. Higher types get weakly higher allocation probabilities, and downward IC constraints bind.

3.2 Two-type random allocation

For $N = 2$, seller profits can be written as

$$\Pi(x_1) = \beta_1 x_1 \theta_1 + \beta_2 v_2^{-1}(x_1 v_2(\theta_1)).$$

There is an interval $(\beta, \bar{\beta})$ such that $x_1^* \in (0, 1)$ if and only if $\beta_2 \in (\beta, \bar{\beta})$.

Interpretation. Risk aversion can make it optimal to allocate randomly to the low type.

3.3 Dynamic implementation with Poisson discounts

For $J = 2$, the seller posts a constant high price p_{n2}^* and random instantaneous discounts to $p_{n1}^* = \theta_{n1}$. The discount arrival rate is

$$\lambda_{n1} = \frac{r x_{n1}^*}{1 - x_{n1}^*}, \quad \frac{\lambda_{n1}}{r + \lambda_{n1}} = x_{n1}^*.$$

Interpretation. Static allocation probability becomes an expected discount factor from waiting for a random sale.

3.4 Sequential discounts for many allocation levels

For $J \geq 3$, the seller uses a stationary Markov price process with states $p_{n1}^*, \dots, p_{nJ}^*$. The high regular state transitions to a first discount, intermediate states transition quickly either to a deeper discount or back to the regular price.

Interpretation. Different types buy at different discount levels while deeper discounts remain risky to wait for.

3.5 Timing of discounts

With observable price histories and two types, the time to the next sale is exponential. With unobserved pre-arrival histories, incentive compatibility is characterized by

$$E[e^{-r(\tau_{1,t}-s)} \mid \tau_{1,t} > s] \geq x_1^*.$$

Interpretation. This permits increasing sale hazards when the absence of a sale is good news that the next sale is closer.

3.6 Loss-aversion version

Gain-loss utility uses $\mu(x) = \lambda\eta x$ if $x \leq 0$ and $\mu(x) = \eta x$ if $x > 0$, with $\lambda > 1$. Under Condition (6), the low type receives random allocation and the high type receives the good for sure.

Interpretation. The mechanism survives when small-stakes risk aversion is interpreted as reference-dependent loss aversion.

4 Theoretical design and credibility

The static optimum is an upper bound. The dynamic price path implements the static allocation probabilities with waiting times. Random timing matters because deterministic discounts create predictable waiting opportunities for high types. The key identifying theoretical force is that the same type should face the same effective price independent of arrival time, while different types are screened by different prices and waiting risks.

5 Tables, figures, and extracted theorem blocks

The paper contains three figures and no empirical tables. The figures and theorem blocks below are directly cropped from the paper.

5.1 Figure 1: Illustration of typical price and quantity patterns

Read: Panels (a) and (b) show two albums with high regular prices and abrupt lower sale prices. Purchases do not obviously collapse immediately before large discounts.

Economic message. The empirical regularity is a high-price mass point plus abrupt, uncertain sales.

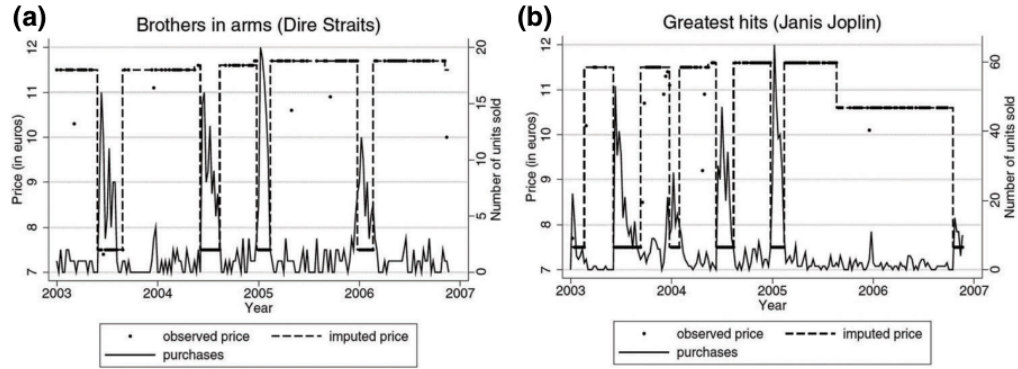


FIGURE 1

Illustration of typical price and quantity patterns (Figure 1 from [Février and Wilner, 2016](#)) for two albums—“Brothers in arms” by Dire Straits, and “Greatest hits” by Janis Joplin—featuring two focal prices (a high price and a discount or sale price). The rights to the material in the figure belong to a third party. *Elsevier*. Appropriate permission must

Figure 1: Illustration of typical price and quantity patterns.

5.2 Figure 2: Numerically determined optimal mechanism for CARA utility

Read: The figure plots p_n^* and x_n^* for 100 evenly spaced types. Allocation probabilities rise with type and are interior for a range of types.

Economic message. Random allocation is robust with many types, not just in a two-type example.

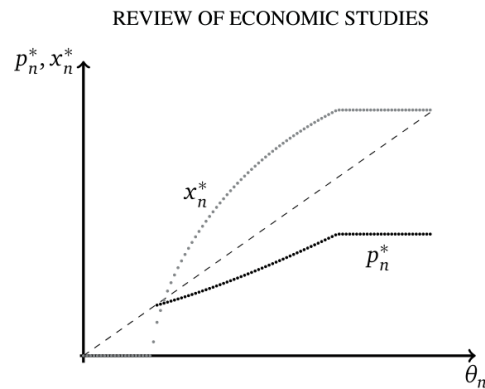


FIGURE 2

ally determined optimal mechanism for a CARA utility function with $R = 1$. We assume that θ_n is evenly spaced between 0 and 9.9, each with the same probability. In particular, $\theta_n = (n - 1)/10$ and

Figure 2: Optimal static mechanism for CARA utility.

5.3 Figure 3: Realizations of random price-path mechanisms

Read: Panel (a) shows a three-type price path with random sequential discounts. Panel (b) shows a many-type price path with repeated short-lived discount episodes.

Economic message. Random allocation probabilities in the static mechanism are implemented as random waiting times for discounts.

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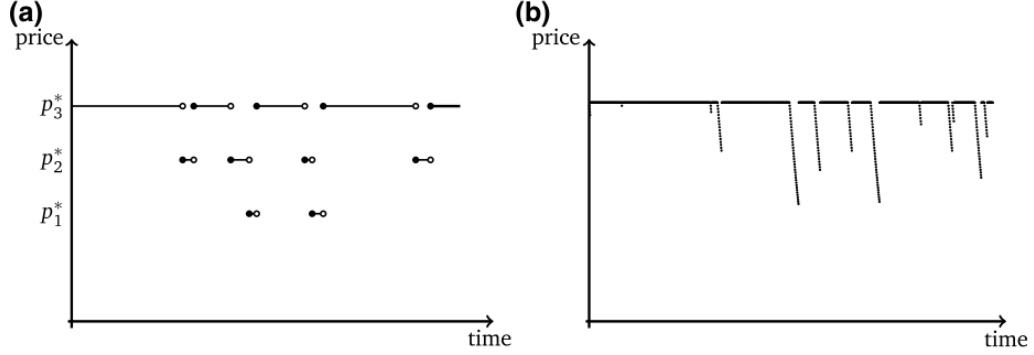


FIGURE 3

(a) Example of a realization of a price-path for $N = 3$ when $0 < x_1^* < x_2^* < x_3^* = 1$. In the example, there are four discount episodes from the regular price p_3^* . In the first and the last, only p_2^* is offered before p_3^* is offered again. In the other two discount episodes a second discount at price p_1^* occurs after the first discount. (b) Example of a

Figure 3: Examples of realizations of optimal random price paths.

5.4 Proposition 1: Unique optimal static mechanism

Read: Proposition 1 states that the optimal mechanism is unique, with increasing allocation probabilities and binding downward IC constraints.

Economic message. The dynamic price path targets a uniquely characterized static screening allocation.

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Proposition 1. *The optimal mechanism is unique. It is fully characterized by a weakly increasing sequence $(x_n^*, p_n^*)_{n=1}^N$ of allocation probabilities and prices for each type such that $x_N^* = 1$. Downward incentive constraints bind: for all $n = 1, \dots, N$, $x_n^* v_n(p_n^*) = x_{n-1}^* v_n(p_{n-1}^*)$, where we put $x_0^* = p_0^* = 0$.*

While Proposition 1 follows closely from the above lemmas, it is worth drawing attention to the proof of uniqueness, which appears new to the literature. Uniqueness is of interest here because it will be important for our discussion of optimal price-paths in the dynamic environment of Section 4.

Optimality of random mechanisms. We do not attempt a full characterization of the optimal allocations $(x_n^*)_{n=1}^N$, but it is important to demonstrate that concavity of the buyer's preferences v_n can imply the optimality of random allocations: that is, it may be that $x_n^* \in (0, 1)$ for some n . We first establish this in the case where $M = 2$ before discussing the case with many types.

Extracted Proposition 1.

5.5 Definition 1: Price-path mechanism

Read: A price-path mechanism consists of a stochastic price process and stopping times for each type and arrival date.

Economic message. This turns the retail pricing problem into dynamic mechanism design.

close to this bound.

Price-path mechanisms: We take a mechanism design approach to defining the seller's choice. This means that we specify not only the random price-path to which the seller commits, but also view the acceptance decision of buyers as being determined by the seller subject to incentive compatibility. We introduce the notation that $\mathcal{B}_A$ represents the Borel sigma algebra on a set $A \subset \mathbb{R}$, while $\mathcal{A}_1 \otimes \mathcal{A}_2$ is the product sigma algebra associated with two sigma algebras $\mathcal{A}_1$ and $\mathcal{A}_2$. Then we have:

Definition 1. A *price-path mechanism* is a pair (P, τ) and a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in \mathbb{R}_+}, \mathcal{P})$ satisfying that:

- (1) $P = (P_t)_{t \in \mathbb{R}_+}$ is a non-negative valued, bounded, and progressively measurable stochastic process defined on $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in \mathbb{R}_+}, \mathcal{P})$, where the filtration $(\mathcal{F}_t)_{t \in \mathbb{R}_+}$ is the one generated by P .
- (2) $\tau = (\tau_{n,t})_{n \in \{1, 2, \dots, N\}, t \in \mathbb{R}_+}$ is a collection of stopping times. For each pair (n, t) , $\tau_{n,t}$ is a stopping time with respect to the given filtration and satisfying that $\tau_{n,t}(\omega) \geq t$ for all $\omega \in \Omega$. A stopping time may be finite or infinite valued. We impose the additional restriction that, for each θ_n , $\tau_{n,t}(\omega)$ is $\mathcal{B}_{\mathbb{R}_+} \otimes \mathcal{F}$ measurable as a function from $\mathbb{R}_+ \times \Omega$ into the non-negative extended real numbers.

A price-path mechanism prescribes a random price-path P , and a buyer acceptance strategy τ , with the acceptance strategy specifying the stopping times $\tau_{n,t}$. The filtered probability space is generally left implicit. A realization of P consists of the path of prices posted by the seller. For each n and t , the realization of $\tau_{n,t}$ is the time at which a buyer of type θ_n arriving at time t

Extracted Definition 1.

5.6 Propositions 6 and 7: Timing of sales

Read: Proposition 6 gives memoryless exponential timing with observable histories. Proposition 7 gives the incentive condition when pre-arrival price histories are unobserved.

Economic message. The theory predicts Poisson-like timing in one information environment and increasing-hazard-compatible timing in another.

specify stopping times for low types, we assume for each ω and each t that either (i) there is a minimum date $s \geq t$ such that the price $P_s(\omega)$ is equal to θ_1 , or (ii) the price is never θ_1 after t . In the first case $\tau_{n,t}(\omega)$ is equal to the minimum date, and in the second $\tau_{n,t}(\omega) = +\infty$. It then makes sense to describe the low type θ_1 as purchasing on the next sales date if there are any future sales.

Optimal price-path mechanisms satisfying the Regularity Conditions exist: the case where sales arrive according to a Poisson process with parameter $\lambda_1 = r x_1^*/(1 - x_1^*)$ is considered above and in the proof of Proposition 4. When sales are optimally determined according to a Poisson process, at any t and irrespective of the history of past prices, the time until the next sale at date $s > t$ is exponentially distributed with parameter λ_1 . Thus, buyers are not able to use the price history to better predict future sales. We can show that this property is in fact shared by any optimal price-path mechanism satisfying the Regularity Conditions in the following sense.

Proposition 6. Consider the model with dynamic arrivals and number of types $N = 2$, and with $x_1^* \in (0, 1)$. Fix any optimal price-path mechanism satisfying the Regularity Conditions. For any date $t > 0$, any positive probability event $A \in \mathcal{F}_t$, the distribution of the time to the next sale is

(a) Proposition 6.

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(We still require that $\tau_{n,t}(\omega) \geq t$ for all $\omega \in \Omega$ as well as $\mathcal{B}_{\mathbb{R}_+} \otimes \mathcal{F}$ measurability of $\tau_{n,t}(\omega)$.) We refer to these mechanisms as *price-path mechanisms with unobserved pre-arrival prices*. Note that now, (P, τ) is *incentive compatible* if, for all n and all t , condition (3) holds for all stopping times $\tilde{\tau}_{n,t}$ with respect to the filtration generated by $P^{\geq t}$ and satisfying that $\tilde{\tau}_{n,t}(\omega) \geq t$ for all $\omega \in \Omega$.

An initial observation is that it remains optimal for the seller to determine sales according to a Poisson process with rate λ_1 . Specifying that high types stop immediately and low types stop at the next sale date, as described above, satisfies the new restrictions on stopping times. The seller obtains what is still an upper bound on expected discounted profits, Π^* .

A range of patterns of sales is now consistent with optimality. The reason is that, because buyers cannot condition their deviations on prices before arrival, the incentive compatibility constraints are easier to satisfy. To understand these patterns, we consider price-path mechanisms with unobserved pre-arrival prices satisfying the Regularity Conditions as well as the requirement that $\mathbb{E}[e^{-r(t_1 - t)}] = x_1^*$ for all t . This last requirement is necessary for obtaining expected discounted profits of Π^* from the cohort arriving at t . We show the following result that provides a condition for incentive compatibility and thus defines a class of price-path mechanisms with

(b) Proposition 7.

5.7 Proposition 8: Loss-aversion extension

Read: Proposition 8 shows that loss aversion can generate random allocation to the low type and sure allocation to the high type.

Economic message. The random-discount mechanism does not rely solely on expected-utility risk aversion.

the scalars $\rho_1 = \rho_1$ for type ρ_1 and $\rho_2 \in (\rho_1, \rho_2)$ for type ρ_2 . For now, these reference points are exogenous; we later argue that these choices can reflect consumer expectations about market prices. We find an optimal static mechanism with random allocation to type θ_1 whenever parameters satisfy

$$\beta_2 \in \left[\frac{\theta_1(1 + \eta)}{\theta_2 + \eta\rho_2}, \frac{\theta_1(1 + \eta + \eta(\lambda - 1))}{\theta_2 + \eta\rho_2 + \theta_1\eta(\lambda - 1)} \right] \quad (6)$$

(an interval of positive length). In particular, we establish the following result.

Proposition 8. *Consider the model of loss aversion with two types and reference points θ_1 for the low type and $\rho_2 \in (\theta_1, \theta_2)$ for the high type. Then if Condition (6) is satisfied, there is an optimal static mechanism in which the low type has a random allocation while the high type receives the good for sure. The prices paid by each type are equal to their reference points.*

To understand the form of the optimal mechanism described in Proposition 8, suppose that Condition (6) holds and the seller offers an optimal direct mechanism as described in Proposition

Extracted Proposition 8.

6 Short summary

- The paper explains random retail price discounts through optimal dynamic price discrimination.
- Risk-averse buyers dislike price risk tied only to arrival time.
- The static optimal mechanism can involve random allocation.
- The dynamic price path implements allocation probabilities through random waiting times.
- Deterministic price paths fail when random allocation is part of the optimum.
- Poisson discount timing appears when price histories are observable; richer timing patterns are possible when pre-arrival histories are unobserved.
- A loss-aversion extension delivers the same basic conclusion.

7 Expanded conclusion

7.1 Core conclusion

Dilmé and Garrett show that random price discounts can be optimal for a seller with commitment power facing forward-looking, risk-averse buyers arriving over time.

7.2 Economic intuition

Random discount timing prevents buyers from predicting low prices while still allowing low and intermediate valuation types to purchase at lower prices. The seller implements static random allocation as dynamic

waiting for random discounts.

7.3 Empirical implications

The model rationalizes high regular prices, abrupt sales, no pre-sale purchase collapse, and discount timing that may be memoryless or increasing-hazard depending on what buyers observe.

7.4 Limitations

The paper is theoretical, assumes commitment, and does not estimate the model structurally. Competition and search frictions may interact with the mechanism in ways left for future work.

7.5 Takeaway

Random sales are not necessarily evidence of myopia or pricing mistakes. They can be an optimal screening device that controls arrival-time price risk and preserves incentives for high-value buyers to purchase at regular prices.